

A fixed-translation obstruction to the uniform Gabor erasure conjecture

Literature-implied status note for arXiv:1611.02540

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Status: literature_implied_answer (full counterexample as written). Conjecture 5.1 of Pfander–Salanevich cannot hold under a cardinality assumption alone. For $\Lambda_M = \{0\} \times \mathbb{Z}_M$, the Gabor system generated by a uniform random unit vector is almost surely a frame and $|\Lambda_M| = M = O(M \log^\alpha M)$ for every $\alpha \geq 0$, but for every fixed retained proportion $p \in (0, 1)$,

$$\Delta_M(p) \leq M \min_j |g(j)|^2 \longrightarrow 0 \quad \text{in probability.}$$

In fact, for $0 < c < 1$, $\Pr\{M \min_j |g(j)|^2 \geq c\} = (1 - c)^{M-1}$.

The conjecture

For a Gabor frame (g, Λ) in $\mathbb{C}^M$, let

$$\Delta(p) = \min_{\substack{\Lambda'' \subset \Lambda \\ |\Lambda''| \geq p|\Lambda|}} \sigma_{\min}^2(\Phi_{\Lambda''}^*), \quad 0 < p < 1.$$

Conjecture 5.1 of Pfander–Salanevich asks for an exponent $\alpha \geq 0$ such that, whenever g is uniform on the complex unit sphere and $|\Lambda| = O(M \log^\alpha M)$, one has

$$\Delta(p) \geq c(p, \varepsilon) \frac{|\Lambda|}{M}$$

with probability at least $1 - \varepsilon$ [1, Section 5, Conjecture 5.1]. As stated, the hypothesis gives only an upper cardinality bound and imposes no spreading condition on Λ .

Idea of the counterexample

Take all modulations but only one translation. The resulting synthesis matrix is a diagonal matrix formed from g times the discrete Fourier matrix. Hence its least frame eigenvalue is controlled by the smallest coordinate of g . A uniform vector on the complex sphere has exponentially distributed coordinate masses after normalization, and its smallest mass is of order M^{-2} , not M^{-1} . Since the full frame itself is one of the subframes allowed in the definition of $\Delta(p)$, no adversarial erasure is even needed.

Exact counterexample

Let

$$\Lambda_M = \{0\} \times \mathbb{Z}_M, \quad \Phi_M = \{M_\ell g : \ell \in \mathbb{Z}_M\},$$

where $(M_\ell g)(m) = e^{2\pi i \ell m/M} g(m)$. If F_M denotes the unnormalized Fourier matrix, then the synthesis matrix is $\text{diag}(g)F_M$. Since $F_M F_M^* = MI$,

$$\Phi_M \Phi_M^* = M \text{diag}(|g(0)|^2, \dots, |g(M-1)|^2),$$

and consequently

$$\sigma_{\min}^2(\Phi_M^*) = M \min_j |g(j)|^2. \quad (1)$$

All coordinates of g are nonzero almost surely, so this is almost surely a Gabor frame. This computation is also the $F = \{0\}$ case of the exact frame bound formula in Salanevich–Strachan, Proposition 4.1 [2, Proposition 4.1].

For a uniform vector on the complex unit sphere, the vector of squared coordinate moduli is uniform on the simplex, i.e.

$$(|g(0)|^2, \dots, |g(M-1)|^2) \sim \text{Dirichlet}(1, \dots, 1).$$

Translating and rescaling the simplex gives, for $0 \leq t \leq 1/M$,

$$\Pr \left\{ \min_j |g(j)|^2 \geq t \right\} = (1 - Mt)^{M-1}.$$

Putting $t = c/M$ yields the exact identity

$$\Pr \left\{ M \min_j |g(j)|^2 \geq c \right\} = (1 - c)^{M-1} \longrightarrow 0 \quad (0 < c < 1). \quad (2)$$

For every $p \in (0, 1)$, the choice $\Lambda'' = \Lambda_M$ is admissible in the minimum defining $\Delta_M(p)$. Equations (1)–(2) therefore imply

$$\Delta_M(p) \leq \sigma_{\min}^2(\Phi_M^*) \longrightarrow 0 \quad \text{in probability.}$$

On the other hand, $|\Lambda_M|/M = 1$. Thus no positive constant depending only on p and ε can give the conjectured lower bound with probability $1 - \varepsilon$ for all sufficiently large M .

The obstruction persists for every fixed number of translations

The same issue is not peculiar to a basis. Fix $K \geq 1$, take $F = \{0, 1, \dots, K-1\}$, and put $\Lambda = F \times \mathbb{Z}_M$. Proposition 4.1 gives the exact lower frame bound

$$A_M = M \min_m \sum_{k \in F} |g(m-k)|^2. \quad (3)$$

Write $g_j = Z_j / (\sum_s |Z_s|^2)^{1/2}$ with $|Z_j|^2$ independent $\text{Exp}(1)$ variables. Among the cyclic K -blocks in (3) are $\lfloor M/K \rfloor$ disjoint blocks, whose unnormalized sums are independent $\text{Gamma}(K, 1)$ variables. Their minimum converges to zero in probability, while $M^{-1} \sum_s |Z_s|^2 \rightarrow 1$. Hence $A_M \rightarrow 0$ in probability for every fixed K . Again $\Delta_M(p) \leq A_M$, whereas the conjectured scale is $|\Lambda|/M = K$.

This also explains the necessary logarithmic threshold in the later paper: for a uniform spherical window and $\Lambda = F \times \mathbb{Z}_M$, its dimension-uniform lower-frame estimate assumes $|F| \geq C \log M$ [2, Example 4.3(iii)].

Scope and repair

The counterexample addresses Conjecture 5.1 exactly as quantified. It does not rule out a repaired theorem requiring both a lower cardinality bound and geometric spreading or randomness of Λ ; nor does it settle the paper’s separate uniform phase-retrieval robustness conjecture. At minimum, $|\Lambda| = O(M \log^\alpha M)$ must be replaced by a two-sided oversampling condition together with structural assumptions that exclude fixed-translation sets. The supporting paper explicitly emphasizes that frame bounds depend on the structure, not merely the cardinality, of Λ [2, Sections 1.2 and 4].

References

- [1] Götz E. Pfander and Palina Salanevich, *Robust Phase Retrieval Algorithm for Time-Frequency Structured Measurements*, arXiv:1611.02540; SIAM J. Imaging Sci. **12** (2019), 736–761.
- [2] Palina Salanevich and Nigel Q. D. Strachan, *Robustness to noise and erasures of Gabor frames in finite dimensions*, arXiv:2509.01325, 2025.